

VLSI Design Interview questions

Below are few of the interesting questions I have gathered from various sources. I will update the questions soon

1. What happens if V_{ds} is increased over saturation?
2. In the I-V characteristics curve, why is the saturation curve flat or constant?
3. What happens if a resistor is added in series with the drain in a mos transistor?
4. What are the different regions of operation in a mos transistor?
5. What are the effects of the output characteristics for a change in the beta (β) value?
6. What is the effect of body bias?
7. What is hot electron effect and how can it be eliminated?
8. What is latchup problem and how can it be eliminated?
9. What is channel length modulation?
10. What is the effect of temperature on threshold voltage?
11. What is the effect of temperature on mobility?
12. What are the different types of scaling?
13. What is stage ratio?
14. What is charge sharing on a bus?
15. What is electron migration and how can it be eliminated?
16. Can both pmos and nmos transistors pass good 1 and good 0? Explain.
17. Why is only nmos used in pass transistor logic?
18. What are the different methodologies used to reduce the charge sharing in dynamic logic?
19. What are setup and hold time violations? How can they be eliminated?
20. Explain the operation of basic sram and dram.
21. Of Read and Write operations, which ones take more time? Explain.
22. What is meant by clock race?
23. What is meant by single phase and double phase clocking?
24. If given a choice between NAND and NOR gates, which one would you pick? Explain.
25. What are stuck-at faults?
26. What is meant by ATPG?
27. What is meant by noise margin in an inverter? How can you overcome it?
28. Why is size of pmos transistor chosen to be close to three times of an nmos transistor?
29. Explain the origin of the various capacitances in the mos transistor and the physical reasoning behind it.
30. Why should the number of CMOS transistors that are connected in series be reduced?
31. What is charge sharing between bus and memory element?
32. What is crosstalk and how can it be avoided?
33. Two inverters are connected in series. The widths of pmos and nmos transistors of the second inverter are 100 and 50 respectively. If the fan-out is assumed to be 3, what would be the widths of the transistors in the first inverter?

34. In the above situation, what would be the widths of the transistors if the first inverter is replaced by NAND and NOR gates?
35. What is the difference between a latch and flip-flop? Give examples of the applications of each.
36. Realize an XOR gate using NAND gate.
37. What are the advantages and disadvantages of Bi-CMOS process?
38. Draw an XOR gate with using minimal number of transistors and explain the operation.
39. What are the critical parameters in a latch and flip-flop?
40. What is the significance of sense amplifier in an SRAM?
41. Explain Domino logic.
42. What are the differences between PALs, PLAs, FPGAs, ASICs and PLDs?
43. What are the advantages of depletion mode devices over the enhancement mode devices?
44. How can the rise and fall times in an inverter be equated?
45. What is meant by leakage current?
46. Realize an OR gate using NAND gate.
47. Realize an NAND gate using a 2:1 multiplexer.
48. Realize an NOR gate using a 2:1 multiplexer.
49. Draw the layout of a simple inverter.
50. What are the substrates of pmos and nmos transistors connected to and explain the results if the connections are interchanged with the other.
51. What are repeaters in VLSI design?
52. What is meant by tunneling problem?
53. What is meant by negative biased instability and how can it be avoided?
54. What is Elmore delay algorithm?
55. What are false and multi cycle paths?
56. What is meant by metastability?
57. What are the various factors that need to be considered while choosing a technology library for a design?
58. What is meant by clock skew and how can it be avoided?
59. When stated as 0.13 μ m CMOS technology, what does 0.13 represent?
60. What is the effect of V_{dd} on delay?
61. What are the various limitations in changing the voltage for less delay?
62. What is the difference between testing and verification?
63. While trying to drive a huge load, driver circuits are designed with number of stages with a gradual increase in sizes. Why is this done so? What not use just one big driver gate?
64. What is the effect of increase in the number of contacts and vias in the interconnect layers?
65. How does the resistance of the metal layer vary with increasing thickness and increasing length?
66. What is the effect of delay, rise and fall times with increase in load capacitance?
67. In a simple inverter circuit, if the pmos in the Pull-Up Network is replaced by an nmos and if the nmos in the Pull-Down Network is replaced by a pmos transistor, will the design work as a non-inverting buffer? Justify your answer.